

Mohamed Fathy Mohamed

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Objective

Results-driven software engineer with proven expertise in backend engineering and problem-solving, seeking to contribute to scalable enterprise-level systems alongside high-performing engineering teams.

Education

Ain-Shams University

Bachelor of Computers and Systems Engineering

Major in Computer and System Engineering.

Grade: Very Good

Relevant Coursework: Database; Software Engineering; Operating Systems; Data Structures and Algorithms; Computer Networks; Distributed systems; Machine Learning; Digital Image Processing; Deep Learning; Quantum networks and Security

Cairo, Egypt

June 2024

Projects

FastLivery (Microservices Delivery Platform)

A scalable delivery ecosystem built with a Microservices architecture, focusing on real-time logistics, location-based services, and secure user management.

- Designed a Backend-for-Frontend (BFF) using **Spring Cloud Gateway**, securing the system with a session-based approach that keeps JWTs server-side to prevent token theft.
- Implemented Distributed Session Management with **Reactive Redis**, ensuring consistent authentication across horizontally scaled gateway instances.
- Integrated OpenRouteService API to calculate precise travel distances and estimated times of arrival (ETA), optimizing route selection based on real-world road networks, Engineered Proximity-Based Search using **Redis Geo-Hashing**, enabling high-performance queries to find the nearest available delivery personnel in real-time.
- Integrated **Keycloak (OIDC/OAuth2)** for centralized Identity Management, handling secure user registration and role-based access control.
- Optimized Service Orchestration using **Eureka Service Discovery** and **Spring Cloud LoadBalancer**, ensuring resilient communication between decoupled services.

Learn Verse(Graduation Project: Excellent Grade)

A robust e-learning platform inspired by Udemy, built to deliver scalable, secure, and high-performance online education.

- Designed and implemented** the backend using **Spring Boot**, ensuring efficient server-side operations and modular scalability.
- Integrated JWT-based authentication and authorization** via **Spring Security**, enforcing secure access control and protecting user data.
- Implemented secure payment processing** using **Stripe**, applying encryption, tokenization, and secure communication protocols to safeguard financial transactions.
- Utilized Spring Data JPA** to manage database interactions, applying best practices in entity design, query optimization, and transactional integrity.
- Structured backend components for maintainability and scalability**, applying clean architecture principles and layered service design.

CPU-Scheduling Operating System

(Apr 2023)

Developed a JavaFX-based desktop app simulating CPU scheduling algorithms. Implemented preemptive and non-preemptive Shortest Job First (SJF) logic and resolved backend bugs to ensure accurate process execution.

Technologies: **Java, JavaFX**

XML Reader

(Jan 2023)

Built a Java-based tool to validate and correct XML files for structural consistency, then extract relational data from social XML inputs. Led the implementation of correction logic and supporting methods.

Mars Rover with autonomous Search and Sampling

(Aug 2022)

Developed a Python-based system to automate the process of search and sampling of rocks on the surface of Mars using python and Image processing algorithms with the highest possible fidelity and terrain covered where the given are the whole world map and sample images

Technologies: **Python, NumPy, CV, Pandas.**

Courses and Competitions

- Ranked 5th on my college in Egyptian Collegiate Programming Contest Qualifications ECPCQ2022
 - Ranked 3rd on my college in Egyptian Collegiate Programming Contest Qualifications ECPCQ2023
 - Ranked 39th out of 200+ teams in Egyptian Collegiate Programming ECPC2023
 - Qualified to ACPC'23 (Arab and Africa collegiate Programming Contest)
 - Qualified to Meta Hacker Cup 2022 round 2
 - Data Structures UCSD (Coursera)
 - Kotlin for Java Developers JetBrains (Coursera)
 - Programming with JavaScript Meta (Coursera)
 - Mathematics for Machine Learning: Linear Algebra Imperial College London (Coursera)
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Technical Skills

- **Languages & Paradigms:** Java, C++, SQL, HTML/CSS, Python, Object-Oriented Programming (OOP), OAuth2.0 / OpenID Connect (OIDC)
- **Frameworks & Libraries:** JavaFX, Spring Core, Spring Boot, Spring Security, JPA, Hibernate, Spring Cloud (Eureka, Config Server, Gateway, Load Balancer), Redis (Geo-hashing & Session Store), Spring Data Redis
- **Tools & Technologies:** Git, GitHub, JUnit 5, Maven, Docker, MySQL, Postgres, Postman, Redis, OAuth2.0 / OpenID Connect (OIDC)

Languages: Fluent in Arabic; Conversational Proficiency in English